AP5041A Vector Signal Generator

Introduction

This configuration guide contains information to help you configure your AP5041A vector signal generator to meet your requirements. Ordering optional capabilities at the time of purchase provides the lowest overall cost of ownership.



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Select Hardware Options

Option	Description	Additional Information
Select Frequency Range (One Required)		
AP5041A-504	Frequency range, 100 kHz to 4 GHz	Type-N (f) RF output connector
AP5041A-506	Frequency range, 100 kHz to 6 GHz	Type-N (f) RF output connector
AP5041A-512	Frequency range, 100 kHz to 12 GHz	Type-N (f) RF output connector
AP5041A-520	Frequency range, 100 kHz to 20 GHz	2.92 mm (m) RF output connector
AP5041A-540	Frequency range, 100 kHz to 40 GHz	2.92 mm (m) RF output connector
Select Close in Phase Noise and Long-Term Frequency Stability Performance Level (One Required)		
AP5041A-LN1	Ultra-low close in phase noise and long-term frequency stability	
AP5041A-LN2	Enhanced ultra-low close in phase noise and long-term frequency stability	
Select Attenuator (Optionally Choose One)		
AP5041A-1E1	Mechanical step attenuator for 4 GHz, 6 GHz, 12 GHz, 20 GHz or 40 GHz	Requires Option 504, 506, 512, 520, or 540, incompatible with Option UNZ
AP5041A-1E2	Electronic step attenuator for 4 GHz frequency range	Requires Option 504
AP5041A-2E2	Electronic step attenuator for 6 GHz, 12 GHz, or 20 GHz frequency range	Requires Option 506, 512, or 520
AP5041A-1EH	Electronic step attenuator and improved harmonics for 6 GHz, 12 GHz, or 20 GHz frequency range	Requires Option 506, 512, or 520
AP5041A-3E2	Electronic step attenuator for 40 GHz frequency range	Requires Option 540
Select Fast Switching (Optional)		
AP5041A-UNZ	Fast switching	Incompatible with Option 1E1
Select Analog Modulation (Optionally Choose Any)		
AP5041A-UNT	AM, FM, phase modulation	
AP5041A-PMR	Pulse modulation	
AP5041A-302	Avionic modulation	
Select Vector System Features (Optionally Choose Any)		
AP5041A-403	Additive white gaussian noise generator	
AP5041A-FCP	Fast control port, external digital I/Q data & PDW streaming	
AP5041A-PCM	Firmware for phase calibratable mode	Requires option PHS, incompatible with Option 540
AP5041A-EXT	External analog AM, FM, IQ inputs	Option UNT required for AM, FM
AP5041A-PDW	Pulse Descriptor Word (PDW)	

Option	Description	Additional Information
Select General Performance Features (Optionally Choose Any)		
AP5041A-PHS	Phase coherent switching	For multi-instrument phase coherency
AP5041A-SNC	Multi-instrument synchronization	Requires PHS
AP5041A-1ER	Flexible reference input	
AP5041A-DC1	24V DC power interface	AC power interface is standard unless option DC1 ordered
AP5041A-GPB	GPIB interface	
AP5041A-006	I/Q waveform data storage on micro-SD card	SD card not included
Select Calibration Documentation (Optional)		
AP5041A-UK6	Commercial calibration certificate with test data	

Select Software Applications

Option	Description
N7600EMBC	Signal Studio for W-CDMA/HSPA+, waveform playback
N7601EMBC	Signal Studio for cdma2000/1xEV-DO, waveform playback
N7602EMBC	Signal Studio for GSM/EDGE/Evo, waveform playback
N7606EMBC	Signal Studio for Bluetooth Classic/EDR/LE, waveform playback
N7608EMBC	Signal Studio Pro for Custom Modulation, waveform playback
N7610EMBC	Signal Studio for IoT, waveform playback
N7611EMBC	Signal Studio for broadcast radio, waveform playback
N7612EMBC	Signal Studio for TD-SCDMA/HSPA, waveform playback
N7615EMBC	Signal Studio for mobile WiMAX, waveform playback
N7617EMBC	Signal Studio for WLAN 802.11, waveform playback
N7623EMBC	Signal Studio for digital video, waveform playback
N7624EMBC	Signal Studio for LTE/LTE-Advanced/LTE-A Pro FDD, waveform playback
N7625EMBC	Signal Studio for LTE/LTE-Advanced TDD, waveform playback
N7626EMBC	Signal Studio for LTE V2X, waveform playback
N7631EMBC	Signal Studio Pro for 5G NR/5G-Advanced, waveform playback
N7632EMBC	PathWave Signal Generation for NR V2X, waveform playback
N7640EMBC	Signal Studio for LMR, waveform playback
Note:	All Signal Studio software application licenses are node-locked and perpetual
IQTools	Free waveform generation software (visit www.keysight.com)

Select Accessories

Option	Description
Y9616A	Rackmount Kit, 2U height
Y9613A	Portable Bag

Recommended Services

Model	Description
R-50C-011-3	Keysight Calibration - Return-to-Keysight - 3 years
R-50C-011-5	Keysight Calibration - Return-to-Keysight - 5 years

Related Literature

AP5041A Vector Signal Generator Product Page

Keysight enables innovators to push the boundaries of engineering by quickly solving design, emulation, and test challenges to create the best product experiences. Start your innovation journey at www.keysight.com.

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